Qingkai Dong

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Education

University of Connecticut Aug 2023 – Present
PhD in Statistics Storrs, CT, USA
Advisors: Prof. HaiYing Wang and Prof. Jun Yan Passed the Ph.D. Qualifying Exam

Zhongnan University of Economics and Law Sep 2020 – May 2023
MS in Mathematical Statistics Wuhan, China
Thesis: Model Averaging and Variable Selection for Accelerated Failure Time Models

Qingdao University Sep 2016 – May 2020
BS in Applied Statistics Qingdao, China

Research Experience

Research Assistant (Statistical Methodology) Aug 2023 – Present
University of Connecticut

- **Primary output:** First-authored methodology paper on rare-feature-aware subsampling for regression models; manuscript in preparation for journal submission.
- Developed theory showing that estimation efficiency for rare binary covariate coefficients is driven by the limited number of rare observations, explaining numerical instability and slow convergence.
- Introduced a balanced subsampling method that quantifies rarity to ensure adequate representation of rare features without pilot sampling, improving robustness and efficiency.
- Implemented the methods in the subsampling R package; validated performance via theoretical results, simulation studies, and real-data applications; produced documentation and reproducible examples.

Research Assistant (Academia–Industry Collaboration) Mar 2024 – Dec 2025
Servier Pharmaceuticals & University of Connecticut

- **Primary output:** Coauthored a manuscript introducing a predictive-modeling–assisted interim analysis framework for censored time-to-event endpoints; manuscript in preparation for journal submission.
- Developed a covariate-informed approach that augments interim data by predicting event times for censored participants, improving estimation of treatment effects and conditional power.
- Proposed evaluation metrics for conditional power forecast accuracy and futility decision quality; conducted extensive simulations to characterize when performance improves or degrades under varied censoring, covariate informativeness/type, prediction range, and sample size.

Research Assistant (Social Determinants of Health, Frailty, and Accelerated Aging in Breast Cancer Survivors)
Jul 2024 – Aug 2024
Departments of Statistics and Human Development and Family Sciences, University of Connecticut

- Helped frame and prepare a study examining social determinants of health, frailty, and accelerated aging among breast cancer survivors.
- Conducted data retrieval, cleaning, and integration across All of Us EHR, survey, and physical-measurement data to support subsequent statistical modeling.

Research Project (Chemical Sensor Data Analytics) Mar 2024 – May 2024
Departments of Statistics and Chemistry, University of Connecticut

- Analyzed fluorescence response data from porphyrinoid sensors; applied clustering and statistical analysis to improve compound differentiation and sensor selection.
- Produced interpretable summaries and visualizations to support experimental decision-making.

Cross-Disciplinary Collaboration (LLM Explainability) 2025 – Present
University of Connecticut & collaborators

- Coauthored a survey manuscript on time series explainability with an emphasis on LLM-enabled semantic explanations; curated benchmarks and an accompanying repository.

Software & Open-Source

R package subsampling (CRAN): A statistical computing toolkit for optimal subsampling in big-data modeling. Supports GLMs, quantile regression, and rare event/rare feature regimes.

- Implemented optimized sampling probability computation targeting variance reduction and prediction accu-

racy, including practical routines for pilot sampling and two-step designs.
- Engineering focus: modular API, documentation/vignettes, reproducible examples, and scalable computation patterns.

Teaching Experience

Principal Instructor Jan 2026 – May 2026
University of Connecticut

- **STAT 3675Q: Statistical Computing.** Independently taught three 50-minute sections serving approximately 20 students; designed course slides and R-based homework assignments, coordinated student presentations, and graded midterm and final examinations.
- Delivered a comprehensive R/RStudio curriculum covering data acquisition and management, visualization, reproducible workflows, regression, nonparametric inference, GLMs, categorical data analysis, time series, and classification.

Academic Presentations

Invited Talk - New England Statistics Symposium (NESS) 2026
Invited Talk - ICSA Applied Statistics Symposium 2026
Poster Presentation - New England Statistics Symposium (NESS) 2026
Poster Presentation - The Design and Analysis of Experiments (DAE) 2026
Poster Presentation - New England Rare Disease Statistics (NERDS) Workshop, Boston, MA Oct 2025
Poster Presentation - Dahshu Data Science Symposium, Storrs, CT Oct 2025

Academic Service

Reviewer: *Computational Statistics and Data Analysis*; *Statistical Papers*; *Sankhya B*; *Journal of Systems Science and Mathematical Sciences (Chinese)*

Awards

UConn Graduate School Conference Participation Award Jul 2026
New England Statistics Symposium (NESS) Poster Award 2026
ICSA Applied Statistics Symposium Student Paper Award 2026
Predoc Fellowship at University of Connecticut 2025
First-class Scholarship at Zhongnan University of Economics and Law 2020, 2022

Technical Skills

Statistics/Methods: survival analysis; interim analysis & conditional power; design simulation; big-data subsampling; model averaging; variable selection

Programming: R (package development), Python (machine learning / deep learning), Git, LaTeX

Selected Publications

Dong Q, Liu B, Zhao H. Weighted Least Squares Model Averaging for Accelerated Failure Time Models. *Computational Statistics and Data Analysis*, 2023.

Zhao H, Dong Q. Variable Selection for Additive Hazards Model with Current Status Data. *Journal of Systems Science and Mathematical Sciences*, 2022.

Zhao H, Liu B, Dong Q. Jackknife Model Averaging of AFT Model with Current Status Data. *Acta Mathematicae Applicatae Sinica*, 2023.

Chen Z, Lucchesi G, Dong Q, Zheng X, Song D, Wen Q, Cheng W, Ni J, Luo D. From Signals to Semantics: A Survey on Time Series Explainability through a Human-Cognitive Lens. *Under review*, 2025.